Final remarks

palm group

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Final remarks

Purpose of the seminar

- Learning about fundamentals of LES modelling
- Learning to apply LES yourself
- To become familiar with the basic structure, numerics and features of a specific LES model (PALM), including its parallelization, i.e. to be able:
  - to install the PALM code
  - to carry out runs with PALM for different scenarios
  - to understand and to analyze the PALM output data
  - to add user-defined code to the model
  - (to debug errors)

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What we did not present

- anelastic approximation
- other turbulence closure schemes (SGS, RANS)
- DNS mode
- aerosol module SALSA
- urban parameterizations (DECEP, SLUrb)
- advanced 3D radiation code (tenstream)
- ocean mode
- 3D-output with masking method, virtual measurement output
- setup / settings for performance optimization
- running PALM on (multiple) GPUs
  - since beginning of 2014, PALM is part of the SPEC ACCEL benchmark (see: http://www.spec.org/accel)
  - ongoing work for openACC and openMP5 porting (offloading)

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More PALM information (I)

- … about the current model status (PALM 6.0)
  - **Maronga, B., and several co-authors, 2020: Overview of the PALM model system 6.0, Geosci. Model Dev., 13, 1335–1372, doi:10.5194/gmd-13-1335-2020.**
  - **Much more papers about specific PALM 6.0 components/modules have recently been published in a special issue of Geosci. Model Dev.**
- … about application for urban flows:
  - Letzel, M.O., M. Krane and S. Raasch, 2008: High resolution urban large-eddy simulation studies from street canyon to neighbourhood scale, Atmos. Env., 42, 8770-8784, doi:10.1016/j.atmosenv.2008.08.001.
  - Gronemeier, T., S. Raasch, and E. Ng, 2017: Effects of unstable stratification on ventilation in Hong Kong, Atmosphere, 8 (9), 168.
- … about non-cyclic boundary conditions:
  - Gryschka, M. and S. Raasch, 2005: Roll Convection During a Cold Air Outbreak: A Large Eddy Simulation with Stationary Model Domain. Geophys. Res. Lett., 32, L14805, doi:10.1029/2005GL022872.
  - Gryschka, M., C. Drüe, D. Etling and S. Raasch. 2008: On the influence of sea-ice inhomogeneities onto roll convection in cold-air outbreaks, Geophys. Res. Lett., 35, L23804, doi:10.1029/2008GL035845.

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More PALM information (II)

- … about the Lagrangian particle model:
  - Steinfeld, G., S. Raasch and T. Markkanen, 2008: Footprints in homogeneously and heterogeneously driven boundary layers derived from a Lagrangian stochastic particle model embedded into large-eddy simulation, Boundary-Layer Meteorol., 129, 225-248.
  - Riechelmann, T., Y. Noh and S. Raasch, 2012: A new method for large-eddy simulations of clouds with Lagrangian droplets including the effects of turbulent collision., New J. Phys., 14, 065008.
  - Hoffmann, F., Y. Noh, and S. Raasch, 2017: The route to raindrop formation in a shallow cumulus cloud simulated by a Lagrangian cloud model, J. Atmos. Sci., 74, 2125-2142.

- … on our webpages:

  http://palm-model.org
  http://docs.palm-model.org

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What to do in case of trouble?

The correct order of actions

1) Carefully check **all** (!) the output.

2) Read the online documentation!

3) Discuss with colleagues (if possible).
   You may use our reddit channel:
   https://www.reddit.com/r/PALM_model_system/

4) See the ticket system under
   http://palm-model.org/trac/wiki/tickets
   if anyone reported a similar problem before.

5) If everything fails, use the ticket system under
   http://palm-model.org/trac/newticket
   and provide information **as precised and as detailed as possible.**

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PALM current & future developments

- immersive boundary condition (realized, but requires special preprocessing tool to generate static drivers)
- improved pressure solver with Neumann boundary conditions (25.10)
- monotone advection scheme to be used with topography
- compressible set of equations
- …

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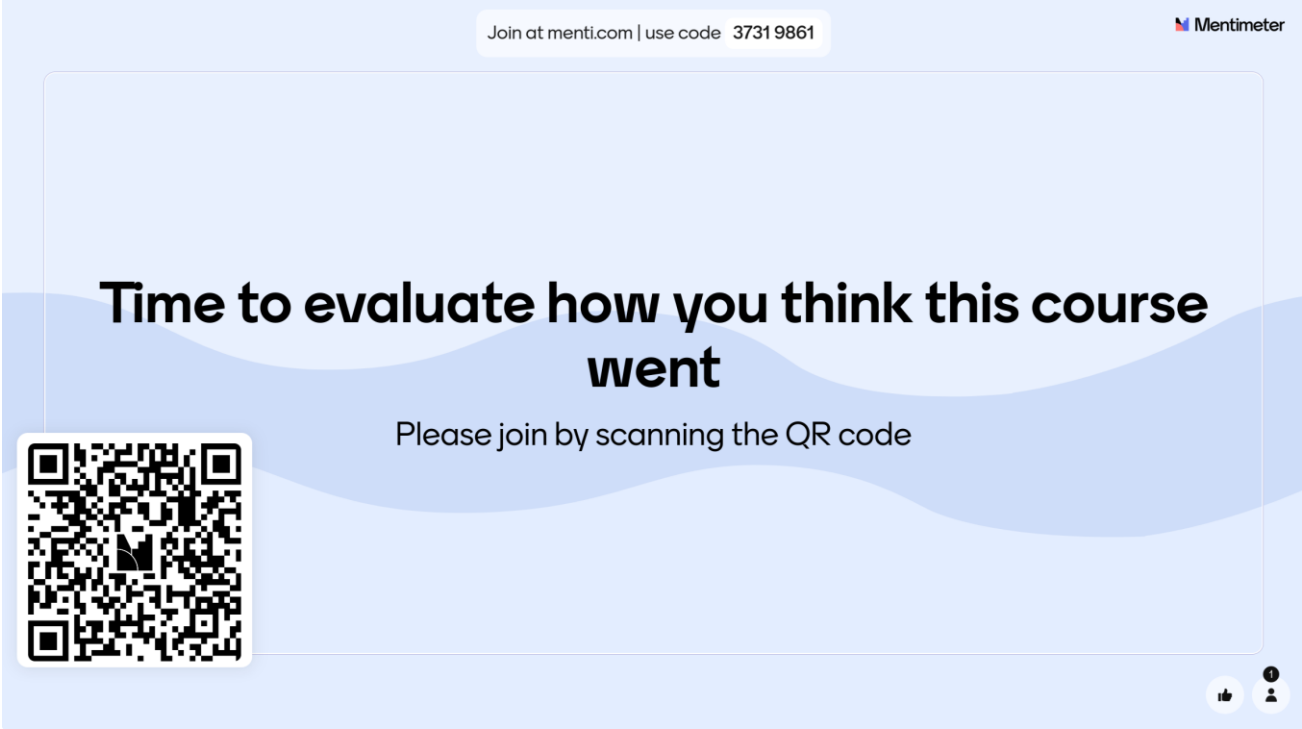
Seminar evaluation

- Please participate the seminar evaluation under link

  https://www.menti.com/alk3up7uxja7?source=qr-instructions-widget

  or use this QR-code:



Participate the annual developer & user conference!

PALM Model Conference 2026 (PMC 2026)

Location: German Meteorological Service (DWD), Offenbach, Germany

Date: 23-27 February, 2026

Registration deadline: …

More information: conference.palm-model.org

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Thank you for participating in this seminar!

Good bye & safe journey!